

BUSINESS ANALYSIS

Project Title

Development of Solar-Powered IoT-Based Automated Feeding Management System for Freshwater Fish Farm

1. Executive Summary

The proposed project aims to address major operational challenges encountered by freshwater fish farmers, particularly in the areas of **feed management, water quality monitoring, energy availability, and remote farm supervision**. Traditional aquaculture practices often depend on manual feeding and periodic water quality inspection, which can result in inconsistent feeding schedules, excessive feed consumption, delayed detection of unfavorable water conditions, increased labor requirements, and inefficient farm management.

The proposed solution is a **solar-powered IoT-based automated feeding management system** composed of two integrated but independently operating prototypes. The first prototype is a **Raspberry Pi-based automated feeding management system** that controls scheduled feeding through a centralized feed storage drum, valve stopper mechanism, feeder/spreader, and PVC distribution tubes connected to multiple feeding endpoints. The second prototype is an **ESP32-based water quality monitoring system with an LTE communication module** that monitors dissolved oxygen (DO), pH, water temperature, and ammonia concentration.

A shared **100-watt solar panel and 12-volt rechargeable battery system** provides renewable energy to support system operation, particularly in fish farms with limited or unreliable access to electricity. A mobile-based monitoring application provides remote access to feeding schedules, feeding records, water quality conditions, battery status, historical data, analytics, and alert notifications.

From a business perspective, the proposed system is positioned as a **farm management and operational efficiency solution** rather than merely an IoT monitoring device. Its primary value is to help fish farmers reduce unnecessary manual work, improve feeding consistency, support timely responses to water quality problems, and utilize renewable energy for system operation.

2. Business Problem Analysis

2.1 Current Business Situation

Freshwater fish farming requires continuous management of several critical activities, particularly:

- fish feeding;
- water quality monitoring;
- energy management;
- farm supervision;
- record keeping; and
- timely response to environmental changes.

In many small- and medium-scale freshwater fish farms, these activities remain highly dependent on manual processes.

Farm workers or owners may need to:

1. manually distribute fish feed;
2. follow feeding schedules using personal judgment or basic reminders;
3. periodically inspect water conditions;
4. manually record water quality readings;
5. monitor equipment and battery conditions;
6. travel to the farm to check operations; and
7. respond to environmental problems after they have already become noticeable.

This creates several operational inefficiencies.

2.2 Identified Business Problems

Problem 1: Inconsistent Feeding

Manual feeding can result in:

- missed feeding schedules;
- irregular feeding intervals;
- inconsistent feed quantities;
- uneven distribution; and
- dependence on worker availability.

These problems may negatively affect fish growth and production efficiency.

Problem 2: Feed Wastage

Overfeeding may cause:

- unnecessary feed expenses;
- uneaten feed accumulation;
- deterioration of water quality; and
- increased operational costs.

Since feed is one of the major expenses in aquaculture, improving feed management can directly contribute to operational efficiency.

Problem 3: Labor Dependency

Manual feeding and monitoring require personnel to be physically present at the farm.

This becomes problematic when:

- the farm is located far from the owner's residence;
 - the farm has multiple ponds;
 - workers are unavailable;
 - feeding must occur at specific times; or
 - environmental conditions change outside regular working hours.
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Problem 4: Delayed Water Quality Detection

Manual water quality testing is often performed periodically rather than continuously.

This creates a risk that changes in:

- dissolved oxygen;
- pH;
- temperature; or
- ammonia

may not be detected immediately.

Delayed detection may result in delayed intervention and potentially affect fish health.

Problem 5: Limited Remote Monitoring

Without an IoT-based system, farmers may need to physically visit the farm to determine:

- whether feeding occurred;

- current water quality conditions;
- battery status;
- system operating condition; and
- whether an abnormal condition exists.

This increases time and transportation requirements.

Problem 6: Dependence on Conventional Electricity

Fish farms in remote areas may experience:

- limited electricity access;
- unstable power supply;
- power interruptions; or
- high electricity costs.

These conditions can affect the reliability of technology-based farm management systems.

3. Business Need

The business need is to develop a system that can provide freshwater fish farmers with a more **efficient, reliable, remotely accessible, and sustainable method of managing feeding and monitoring pond conditions**.

The system should provide the following business capabilities:

Feeding Management

- scheduled automated feeding;
- consistent feed dispensing;
- centralized feed storage;
- multi-point feed distribution;
- reduced dependence on manual feeding.

Water Quality Management

- continuous monitoring;
- remote access to water quality data;
- historical data recording;
- graphical analysis;
- abnormal-condition detection.

Energy Management

- solar-powered operation;
- rechargeable battery backup;
- battery status monitoring;
- reduced dependence on grid electricity.

Remote Farm Management

- mobile monitoring;
- feeding schedule management;
- real-time notifications;
- historical records;
- data-driven decision support.

4. Proposed Business Solution

The proposed solution is a **Solar-Powered IoT-Based Automated Feeding Management System** designed specifically for freshwater fish farm operations.

The system consists of two main prototypes.

Prototype 1: Automated Feeding Management

The Raspberry Pi controls the automated feeding process.

The feeding process is:

Central Feed Storage Drum

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Valve Stopper Mechanism

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Feeder/Spreader

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PVC Distribution Tubes

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Multiple Feeding Endpoints

The Raspberry Pi executes programmed feeding schedules and records feeding activities.

Prototype 2: Water Quality Monitoring

The ESP32 serves as the dedicated monitoring controller.

The system monitors:

- Dissolved Oxygen;
- pH;
- Water Temperature;
- Ammonia.

The ESP32 is connected to an LTE communication module to transmit monitoring information to the cloud platform.

Solar Energy System

The energy system consists of:

- 100-watt solar panel;
- solar charge controller;
- 12-volt rechargeable battery;
- power regulation and distribution components.

The solar system provides renewable energy to support the operation of both prototypes.

Mobile-Based Application

The application provides:

- feeding schedule management;
- feeding activity monitoring;
- water quality monitoring;
- battery monitoring;
- historical records;
- data analytics;

- alert notifications.

Alerts may be delivered through:

- sound;
- vibration;
- messages or notifications.

5. Stakeholder Analysis

Stakeholder	Role	Business Interest	Expected Benefit
Fish Farmers	Primary users	Efficient management	farm Reduced labor and improved monitoring
Farm Owners	Decision makers	Productivity cost control	and Better operational decisions
Farm Workers	System operators	Easier operations	daily Reduced manual workload
Researchers/Students	Developers and evaluators	Technology development	Research and technical knowledge
Fisheries Organizations	Industry stakeholders	Sustainable aquaculture	Potential technology adoption
Local Government Units	Community support	Sustainable livelihood programs	Possible support for technology adoption
Future Developers	System Technology stakeholders	System improvement	Foundation for future enhancements

The **primary stakeholder** is the freshwater fish farmer because the system is designed to address the farmer's operational challenges.

6. Business Value Proposition

The system's value proposition can be summarized as:

"A solar-powered, IoT-enabled farm management solution that automates scheduled fish feeding while providing remote monitoring of critical water quality conditions, battery status, and farm operations through a mobile-based application."

The proposed system provides value through five major areas.

1. Operational Efficiency

Automated feeding reduces the need for continuous manual intervention.

2. Cost Efficiency

Improved feed management may reduce unnecessary feed consumption and labor requirements.

3. Environmental Monitoring

Continuous monitoring helps identify unfavorable water conditions earlier.

4. Energy Sustainability

Solar energy reduces reliance on conventional electrical power.

5. Decision Support

Historical data and analytics help farmers identify trends and make better operational decisions.

7. Business Process Analysis

Current Process

Feeding

Farmer/Worker

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Manually prepares feed

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Travels to pond

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Manually distributes feed

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Records activity manually or not at all

Water Monitoring

Farmer/Worker



Collects water sample



Manually checks parameters



Records readings



Interprets results



Takes action if necessary

Proposed Process

Automated Feeding

Farmer



Sets Feeding Schedule



Raspberry Pi



RTC/Scheduling Mechanism



Valve Stopper



Feeder/Spreader



PVC Distribution System



Multiple Feeding Points



Feeding Activity Recorded



Mobile Application Updated

Water Quality Monitoring

Water Quality Sensors



ESP32



Data Processing



LTE Communication



Cloud Platform



Mobile Application



Analytics



Alert Notification



Farmer Decision/Action

This represents a shift from **reactive and manual farm management** toward **automated and data-supported farm management**.

8. SWOT Analysis

Strengths

- Solar-powered operation.
- Automated feeding schedules.
- Centralized feed storage.
- Multi-point feed distribution.
- Continuous monitoring of four important water quality parameters.
- LTE-based remote monitoring.
- Mobile-based system access.
- Historical data and analytics.
- Battery monitoring.
- Real-time alerts.
- Two dedicated prototypes with specialized controllers.

Weaknesses

- Initial development cost.
- Dependence on sensor accuracy.
- Solar power depends on sunlight availability.
- LTE connectivity may vary by location.
- Requires periodic hardware maintenance.

- Sensors require calibration.
- Mechanical feeding components may require cleaning and maintenance.
- Users may require basic technical knowledge.

Opportunities

- Expansion to multiple fish ponds.
- Integration of additional water quality sensors.
- AI-based feeding optimization in future versions.
- Predictive water quality analytics.
- Commercialization for small and medium fish farms.
- Integration with fisheries management programs.
- Development of a larger-scale commercial version.
- Integration with additional renewable energy sources.

Threats

- Extreme weather conditions.
 - Sensor degradation.
 - Equipment failure.
 - LTE network outages.
 - Solar panel damage or theft.
 - Battery degradation.
 - Increasing hardware costs.
 - Resistance to technology adoption.
 - Cybersecurity risks.
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9. Feasibility Analysis

9.1 Technical Feasibility

The project is technically feasible because the required components are commercially available and suitable for IoT applications.

The proposed architecture uses:

- Raspberry Pi 4;
- ESP32;
- LTE communication;
- water quality sensors;
- solar energy components;
- cloud infrastructure;
- mobile-based application.

The two-prototype architecture also allows each subsystem to perform its primary function independently.

Assessment: Feasible

9.2 Operational Feasibility

The system is operationally feasible because it is designed around the daily activities of fish farmers.

The system reduces manual tasks while maintaining user control over:

- feeding schedules;
- monitoring;
- data interpretation;
- alerts;
- farm management decisions.

The mobile-based interface also allows the farmer to monitor the system without always being physically present at the farm.

Assessment: Feasible with user training

9.3 Economic Feasibility

The system requires an initial investment for:

- controllers;
- sensors;
- mechanical components;
- solar equipment;
- battery;
- communication equipment;
- software infrastructure.

However, potential long-term benefits include:

- reduced labor requirements;
- reduced feed wastage;
- reduced dependence on grid electricity;
- reduced travel for manual monitoring;
- improved response to water quality changes.

The economic feasibility should ultimately be validated through a **cost-benefit analysis and return-on-investment assessment** during future commercial deployment.

Assessment: Potentially feasible

9.4 Schedule Feasibility

The project can be completed within an academic development cycle if implementation is divided into manageable phases:

1. Requirements analysis
2. System design
3. Hardware procurement
4. Feeding prototype development
5. Water monitoring prototype development
6. Solar power integration
7. Backend development
8. Mobile application development
9. System integration
10. Testing
11. Evaluation
12. Documentation

Assessment: Feasible with controlled project management

10. Risk Analysis

Risk	Probability	Impact	Mitigation
Sensor malfunction	Medium	High	Calibration and replacement testing
LTE failure	Medium	High	Local buffering and automatic reconnection
Low solar generation	Medium	High	Battery monitoring and power budgeting
Battery degradation	Medium	Medium	Battery health monitoring
Feeding mechanism blockage	Medium	High	Mechanical inspection and maintenance
Feed wastage	Medium	Medium	Feeding accuracy testing
Incorrect sensor readings	Medium	High	Calibration against reference instruments

Risk	Probability	Impact	Mitigation
Internet/cloud outage	Medium	High	Local data storage and synchronization
Hardware damage	Low-Medium	High	Waterproof enclosure and protection
User difficulty	Medium	Medium	Simple interface and user training
Cybersecurity threats	Low-Medium	High	Authentication and secure communication

11. Key Performance Indicators (KPIs)

To determine whether the project successfully delivers business value, the following KPIs should be monitored:

Feeding Management

- Feeding accuracy percentage
- Successful scheduled feeding rate
- Feed distribution consistency
- Number of missed feeding events

Water Quality Monitoring

- DO measurement accuracy
- pH measurement accuracy
- Temperature measurement accuracy
- Ammonia measurement accuracy

Communication

- Data transmission success rate
- Average communication latency
- Data synchronization success rate

Solar Energy

- Battery operating duration
- Solar charging time
- Battery status accuracy
- System runtime under low sunlight

Application

- Alert delivery success rate

- Application response time
- Data visualization availability
- User task completion rate

User Evaluation

- Functionality
 - Reliability
 - Usability
 - Performance efficiency
 - User satisfaction
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12. Business Impact

The expected business impact of the system can be categorized into four areas.

Operational Impact

The system reduces repetitive manual feeding and monitoring tasks, allowing farmers to focus on higher-value farm management activities.

Financial Impact

Better feeding management may reduce feed wastage, while solar power can reduce dependence on conventional electricity.

Environmental Impact

Improved feeding control may reduce uneaten feed accumulation, while continuous monitoring supports better management of pond conditions.

Strategic Impact

The project introduces a pathway toward **precision aquaculture**, where farm management decisions are supported by real-time data, automation, and analytics.

13. Project Success Criteria

From a project management perspective, the project should be considered successful if it achieves the following:

1. The automated feeding prototype successfully performs programmed feeding schedules.
 2. The feeding mechanism achieves an acceptable level of dispensing accuracy.
 3. The water monitoring prototype accurately measures DO, pH, temperature, and ammonia based on defined evaluation criteria.
 4. The ESP32/LTE subsystem successfully transmits monitoring data to the cloud platform.
 5. The solar-powered system provides sufficient energy to support the intended operating period.
 6. The mobile application successfully provides monitoring, feeding management, analytics, battery status, and alert notifications.
 7. The system continues essential local operations during temporary connectivity interruptions.
 8. The developed system receives acceptable evaluations for functionality, reliability, usability, performance efficiency, and user satisfaction.
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14. Recommended Business Analysis Conclusion

The proposed project is **technically and operationally viable as an academic prototype and has potential for future commercialization**. Its strongest business value lies in addressing several interconnected problems in freshwater fish farming rather than solving only one operational issue.

The integration of **automated feeding, water quality monitoring, solar energy, LTE communication, mobile monitoring, data analytics, and real-time alerts** creates a more comprehensive farm management solution. The use of two dedicated prototypes also provides a modular architecture in which the feeding and monitoring functions can operate independently while sharing centralized cloud services and energy resources.

However, before the system can be considered commercially deployable, further validation is necessary in terms of **long-term reliability, cost-benefit performance, sensor durability, solar energy autonomy, mechanical feeding reliability, connectivity stability, cybersecurity, and large-scale farm deployment**.

Therefore, from a project management perspective, the project should be positioned as a **prototype-level smart aquaculture management solution** that demonstrates the feasibility of integrating automation, IoT, renewable energy, and remote data management into freshwater fish farming. Its immediate success should be measured not by full commercial replacement of existing farm operations, but by its ability to **demonstrably improve feeding consistency, monitoring accessibility, operational efficiency, and sustainable energy utilization** within the defined project scope.